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POWERING DELHI
EMPOWERING CONSUMERS

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Technological Innovation for Loss Reduction

Powering Delhi, Empowering Consumers

BSES Yamuna Power Limited

BSES: 23 years of serving the National Capital

PROPOSED TARGETS FOR TECHNICAL LOSS LEVELS
Sub- Transmission & Distribution System

S#	System component	Prescribed Loss Levels (%)		BYPL Loss % (Actual)
		Target Level (%)	Max. Tolerable Limit (%)	
1	Sub-transmission system(66kV, 33 kV)	1.5%	2%	1%
2	Primary distribution system (11 kV)	1.5%	2.5%	1.5%
3	Distribution LT lines & Service connections (415V)	2.5%	3%	2.5% -3%
	TOTAL	**5.50%**	**8.00%**	5% - 5.5%

Source - CEA's Electricity Distribution Network Planning Criteria 2023

Major Reasons For High Technical Losses

Disproportionate Investment In T&D

- From 1956 -97 : investment allocation across Power value Chain → G: T+D :: 1: 0.45 against the recognized 1:1

- Low investment has resulted in overloading of the distribution system.

1. Haphazard network growths due to **Short-term Objective** of extension of power supply to new areas.

2. **Rural Electrification Through Long Feeders -** 11kV / LT lines.

3. Too many stages of **Transformation Stages.**

4. **Improper Load Management.**

5. **Inadequate Reactive Compensation**

6. **Poor Quality Equipment -** Agricultural pumps, coolers, air conditioners & other industrial loads

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Effort Required

- **Smart metering** for accurate energy audit
- Establishing **Grid Stations & Sub-stations near to the load centre.**
- **Reactive Power Compensation** through APFC across all voltage levels & near to the customer load
- Conversion of existing non star rated equipment (incl. X'mers) to **BEE's Star rated equipment**
- Feeder Phase Current and Load Balancing with **Distribution Automation & ADMS deployment**
- **Optimising feeder lengths** to minimise feeder losses & conversion to underground
- Replacement of **Overhead Feeders with Cables**
- **AI/ML based energy audit and data analytics for theft analysis & case booking.**

Effort Required

System component	Loss Reduction Measure	Challenges
Sub-transmission system (66kV, 33 kV)	Establishing Grid Stations near load centre.	Space constraint; High Cost – Capex & Land Lease
	Reactive Power Compensation through APFC	High Capex
	Conversion / Installation of BEE's Star rated equipment incl Transformers	High Capex ; Long payback period
Primary distribution system (11 kV)	Establishing Sub-stations near load centre.	Space constraint; High Cost – Capex & Land Lease
	Reactive Power Compensation through APFC	High Capex
	Feeder Phase Current and Load Balancing with Distribution Automation & ADMS deployment	High Capex
	Conversion / Installation of BEE's Star rated equipment incl Transformers	High Capex ; Long payback period
Distribution LT lines & Service connections (415V)	Optimising long LT feeder lengths to minimise feeder losses & conversion to underground	Space constraints for network expansion in urban / unauthorised areas
	Feeder Phase Current and Load Balancing with Distribution Automation & ADMS deployment	High Capex ; Long payback period
	Replacement of Overhead Feeders with Cables	High Capex ; Long payback period
	AI/ML based energy audit and data analytics for theft analysis & case booking.	GIS linked Smart Metering for segregating Direct Theft (DT) from discoms LT technical losses through efficient energy audit.

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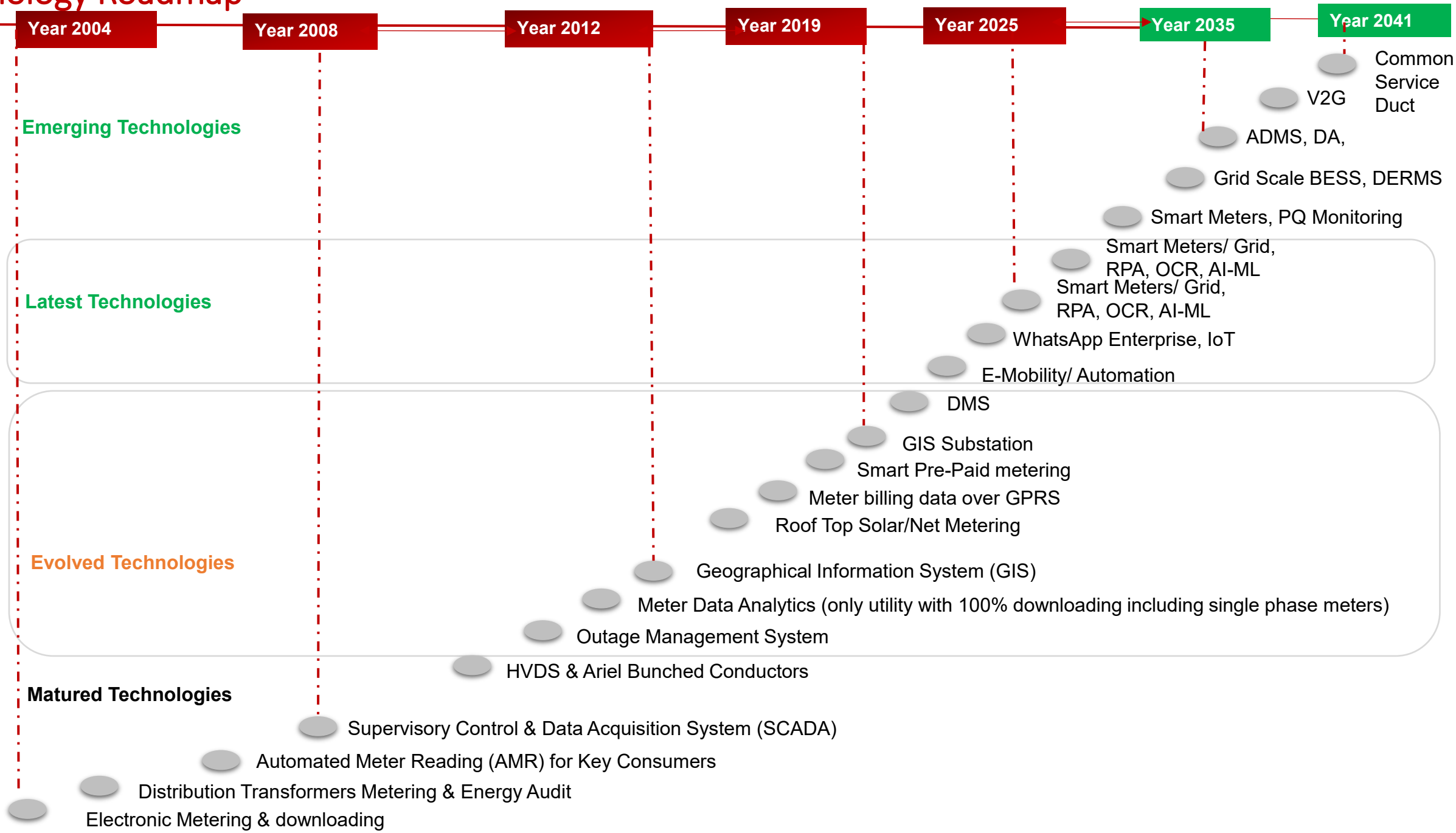
Technology Roadmap



Technology Innovation

Project	**E House Grid (India's 1st)**
Brief Description	• E-House Grid Substation - 36 MVA, 33/11kV, 33kV GIS Switchgear. 11kV AIS Switchgear. • Deployed at C-Block Krishna Nagar. • Partner : Siemens
Benefits	• 44% Space saving (foot print - 895 sq.mt Vs conventional Grid S/stn size 1,600 sq.mt) • Faster Execution with minimum Civil Works. • Improved Safety, power reliability & better aesthetics.

Project	**Hybrid Switchgear (1st for any Indian Discom)**
Brief Description	• Compact, GIS 66kV switchgear combining CT, Isolator & Switchgear units into single unit. • Remote Operation of Isolators through SCADA • Installed in Ghonda, Nand Nagri, Bhagirathi & Yamuna Vihar Grid • Partner : ABB
Benefits	• Back to Back Bay addition leading to space savings of upto 55% • Useful in congested grids requiring augmentation / bay addition • Improved Safety, power reliability & better aesthetics.

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Technology Innovation

Project	**Micro Sub-Station (India's 1st)**
Brief Description	• Touch proof, compact electrical substation with Oil type Transformer. • 11kV / 415 V; DT- 1MVA; RMU-3Way;LT panel- I/C- 1No, O/G-6 no's • Installed in Daryaganj, Shankar Rd divisions • Partner : Lucy Electric
Benefits	• 84% space saving (foot print - 5 sq.mt Vs conventional S/stn size 30 sq.mt) • Useful in urban areas with space limitation. • Improved Safety, power reliability & better aesthetics.

Project	**Smart PSS with N-1 Redundancy**
Brief Description	• Touch proof, compact electrical substation with Dry type Transformer. • 11kV / 415V; DT- 1.6 MVA; LT Incomer, Bus-coupler- 3200A; O/g- 1600A- 2nos & 630A- 2nos • Redundancy at LT side – LT supply get restore with adjacent substation. • Installed in Daryaganj divisions • Partner : Siemens
Benefits	• Temperature health monitoring • Distribution Automation with enhanced power reliability, safety.

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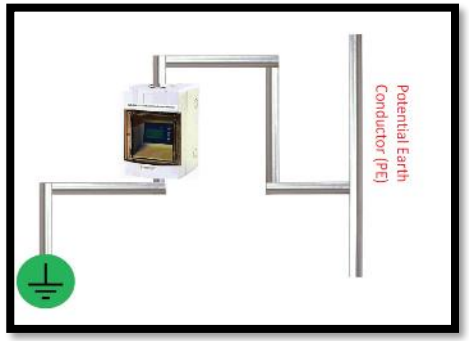
Technology Innovation



Project	Intelligent Grid Earth Resistance Monitoring (India's 1st)
Brief Description	• Consists of Voltage Generator, Current Sensing Coils & Display Screen • Remote monitoring and data logging through SCADA • Deployed at IP Extension Grid Substation • Partner : Schneider Electric
Benefits	• Intelligent, continual monitoring leaving no room for errors • Real time monitoring eliminating the probability of undetected failures • Improved reliability with lower breakdown & network downtime.

Project	Smart Lightning Arrestor
Brief Description	• Remote health monitoring of Lightening Arrestor (LA) parameters – leakage current & Lightening Strike counter • Interoperable & scalable solution as it is based on MODBUS open protocol. • Installed in GT Road Grid & Yamuna Vihar Grid • Partner : Oblum Industries
Benefits	• Useful in urban areas with space limitation. • Improved Safety, power reliability & better aesthetics.

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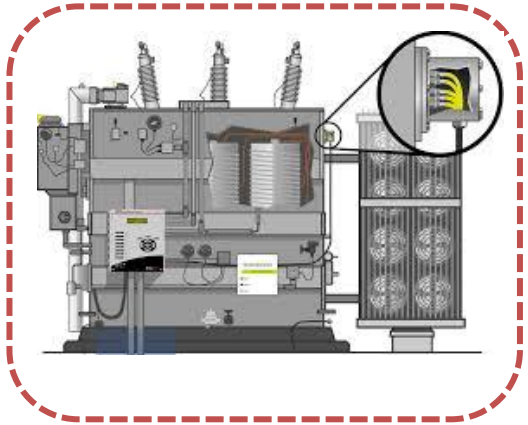
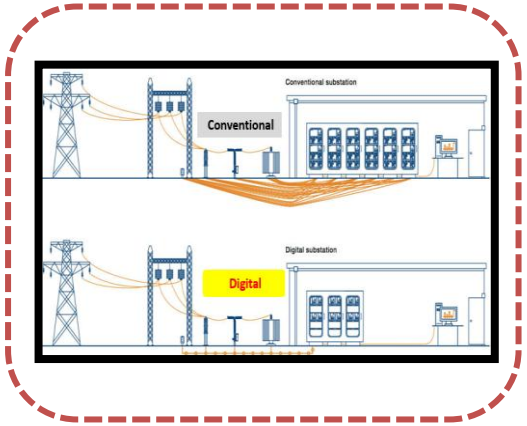
Technology Innovation



Project	Hydrogen Gas Monitoring System for Power Transformer
Brief Description	• Provides 24X7 data of hydrogen gas concentration in the transformer oil which enables early detection of failures and prevent any major outage. • Installed in IP Extension Grid. • Partner : Schneider Electric
Benefits	• Enhanced Reliability - Prevention of unplanned downtimes by early fault detection. • Higher Operational Efficiency - Efficient scheduling of maintenance work • O&M Cost Optimization - Prevention of major damages to power transformer due to early fault detection



Project	Digital Substation based on IEC 61850
Brief Description	• Solution is based on International Protocol IEC 61850 hence it is completely interoperable • Only Optical Fiber Cable is laid from the outdoor yard to the grid control room • Control cable trench not required • Installed in Line bay at Sonia Vihar Grid. • Partner : General Electric
Benefits	• Quick Testing and Commissioning with easy troubleshooting • No Fire Hazard • Improved & reliable Grid level communication.

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Project	**Insulated Dropout Fuse in HVDS Area**
Brief Description	• Insulating 11 kV vulnerable HVDS point in highly populated/encroached' HVDS colonies. • Insulating 1Ph DO fuse units with insulation cover made of glass filled nylon with 3mm thickness & breakdown strength of 10 kV/mm • Installed in HVDS areas of Mayur Vihar Ph-III , Yamuna Vihar. • Partner : GK Electricals
Benefits	• Consumer safety of people residing in HVDS network • Prevents nuisance tripping thereby improving reliability.

Project	**Intelligent Group Metering System (IGMS)**
Brief Description	• Pole Mounted group energy metering solution with home display unit at consumer premises • Communication with utility server is on Cellular with no scalability constraint • Communication between Meter & Home Display unit uses proven RF/Bluetooth • Installed in HVDS areas of Mayur Vihar Ph-III , Yamuna Vihar. • Partner : Genus Meter
Benefits	• Meter Out of Reach from Consumer as it is placed on Pole • Theft Control as only home display unit is deployed at consumer premises. • Drastic AT&C loss reduction shows that iGMS is a potent tool to tackle power theft.

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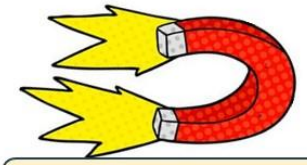
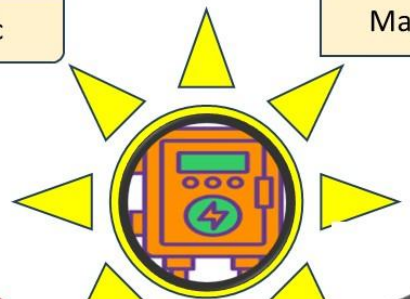
Energy Meters

~100% meters (incl 1 Ph meters) downloading since >15 years

Upto 35 kVc

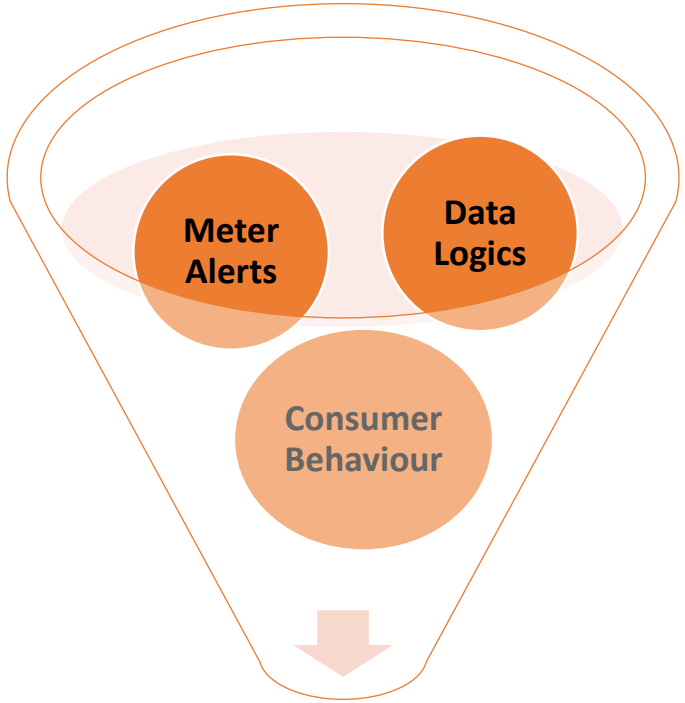


Magnetic Tamper



CT Open / Bypass

EEPROM Tampering / Meter Cloning



Analytic Lead Generations

- Tamper Features -

- Meter Data Analytic Engine -

Thank You

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